외계행성 대기
Exoplanetary Atmospheres

Lecture 7: Vertical Thermal Structure of a Planetary Atmosphere
2026 October 26 (Monday)

Based on: Chapters 6 and 9 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 4 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 7

Approximate Solutions: Rosseland, Eddington, and the Grey Atmosphere

Why the Vertical Structure Matters

Hydrostatic Equilibrium and the Scale Height

Surface Temperature in Simple Greenhouse Models

Convection versus Radiation

The Radiative-Equilibrium Temperature Profile

Greenhouse and Anti-Greenhouse Effects

The Adiabatic Temperature Profile

The One-Dimensional T–P Profile

Temperature Retrieval

Summary

Approximate solutions

How to solve the radiative transfer equation:

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\alpha_\nu^{\mathrm{ext}}\,(I_\nu - S_\nu), \qquad S_\nu = (1-\varepsilon_\nu)J_\nu + \varepsilon_\nu B_\nu, \quad \text{with} \quad \varepsilon_\nu = \frac{\alpha_\nu^{\mathrm{abs}}}{\alpha_\nu^{\mathrm{ext}}}.$$

(Recall: $S_\nu \equiv \varepsilon_\nu^{\mathrm{tot}}/\alpha_\nu^{\mathrm{ext}}$, where the total emissivity is thermal emission, $\alpha_\nu^{\mathrm{abs}} B_\nu$ in LTE (Kirchhoff), plus scattering into the beam, $\alpha_\nu^{\mathrm{scat}} \oint \mathscr{P}(\Theta) I_\nu \,\mathrm{d}\Omega'$ with $\oint \mathscr{P}\,\mathrm{d}\Omega = 1$, which for isotropic ($\mathscr{P} = 1/4\pi$) and coherent scattering is $\alpha_\nu^{\mathrm{scat}} \oint I_\nu \,\mathrm{d}\Omega'/4\pi = \alpha_\nu^{\mathrm{scat}} J_\nu$ (Lecture 3); dividing by $\alpha_\nu^{\mathrm{ext}} = \alpha_\nu^{\mathrm{abs}} + \alpha_\nu^{\mathrm{scat}}$ gives the weights ε_ν and $1-\varepsilon_\nu$.)

(Here $\varepsilon_\nu = 1 - \omega_0$: the fraction of extinction that is true absorption, so $S = (1-\omega_0)B + \omega_0 J$ as before.)

We will learn two approximations to solve the equation in a **plane-parallel** medium, primarily developed for stellar atmospheres:

- the **Rosseland** approximation (radiative diffusion),
- the **Eddington** approximation.

Together they give the temperature structure of a *grey* atmosphere and the limb darkening of the Sun, and they lead to the Milne solution.

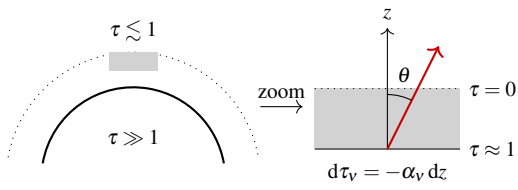
Plane-parallel atmosphere

Stellar (and planetary) atmosphere.

- The atmosphere of a star (or planet) is its outer region, which determines the properties of the radiative flux that emanates from (are incident on) its surface.
- Computing its structure needs the detailed monochromatic opacity and powerful numerical codes; but with appropriate approximations interesting results follow analytically.

Plane-parallel atmosphere.

- We ignore the curvature of the underlying body (frequency independent opacity). Suitable when the atmosphere is a thin layer, "locally virtually flat".
- The optical depth is a function of the vertical distance z rather than the distance s along the ray; it increases downward, with $\tau = 0$ at the top: $\mathrm{d}\tau_\nu = -\alpha_\nu \, \mathrm{d}z$.



(1) Radiative diffusion: the Rosseland approximation

Imagine a plane-parallel medium in which ρ and T depend only on the depth z. Along a ray, $\mathrm{d}s = \mathrm{d}z/\cos\theta = \mathrm{d}z/\mu$, so

$$\mu \frac{\partial I_\nu(z,\mu)}{\partial z} = -\alpha_\nu^{\mathrm{ext}}(I_\nu - S_\nu) \quad \Longrightarrow \quad I_\nu(z,\mu) = S_\nu - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial I_\nu}{\partial z}.$$

- **"Zeroth" approximation.** Deep in the material, *all quantities change slowly on the scale of a mean free path* $\ell = 1/\alpha^{\mathrm{ext}}$, so the derivative term is very small:

$$I_\nu^{(0)}(z,\mu) \approx S_\nu^{(0)}(T) \quad \therefore \quad J_\nu^{(0)} = S_\nu^{(0)} \text{ and } I_\nu^{(0)} = S_\nu^{(0)} = B_\nu \quad (\mathrm{LTE}),$$

  independent of the angle.

- **"First" approximation.** Insert $I^{(0)}$ in the derivative term:

$$I_\nu^{(1)}(z,\mu) \approx S_\nu^{(0)} - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial I_\nu^{(0)}}{\partial z} = B_\nu(T) - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu(T)}{\partial z} \quad \to \text{ linear in } \mu.$$

  With $\mathrm{d}\tau_\nu = -\alpha_\nu^{\mathrm{ext}}\mathrm{d}z$ this is $I_\nu \simeq B_\nu + \mu\,\mathrm{d}B_\nu/\mathrm{d}\tau_\nu$.

What the first approximation means

$$\boxed{I_\nu(\tau_\nu,\mu) \simeq B_\nu(\tau_\nu) + \mu\,\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}} \qquad (\tau_\nu \text{ increasing inward})$$

- The leading term is **isotropic and Planckian**, $I_\nu \simeq S_\nu \simeq B_\nu$; as $\tau \to \infty$, $I = S = B$. A small anisotropy $\propto \mu$ carries the net flux.
- Its moments give $K = J/3$ *exactly* — the Eddington relation of the next part, and the **third Eddington coefficient** $\varepsilon_3 = K/J = 1/3$ (Heng's textbook). In the two-stream picture the up- and down-going streams merge here: net and total fluxes are no longer distinct, and scattering is effectively isotropic ($g_0 = 0$). (see next page)
- This is a **deep-interior** approximation, not a solution for the escaping radiation at $\tau_\nu \lesssim 1$.
- Large *scattering* depth alone does not guarantee thermalization: $S_\nu \to B_\nu$ requires enough true absorption ($\tau_* = \sqrt{3\varepsilon}\,\tau \gg 1$, below).

Seager's Eq. (6.49) prints $I \approx B - \mu\,\mathrm{d}B/\mathrm{d}\tau$; with τ increasing inward the sign must be $+$ to agree with the transfer equation and with her Eq. (6.51).

Why the two streams merge deep inside

Deep inside, $I_\nu \simeq B_\nu + \mu\, \mathrm{d}B_\nu/\mathrm{d}\tau_\nu$ is almost isotropic. Two simplifications of the two-stream picture follow.

- **Net and total fluxes are no longer distinct.** The two hemispheric fluxes are

$$F_\pm = 2\pi \int_0^1 I_\nu(\pm\mu)\, \mu\, \mathrm{d}\mu \simeq \pi B_\nu \pm \frac{2\pi}{3} \frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}.$$

  Their sum, the total flux $\simeq 2\pi B_\nu$, is fixed by the local temperature and carries no new information. Only the small difference, the net flux $F_+ - F_-$ (from the μ term), matters. One flux and J suffice; two separate streams are not needed.

- **Scattering is effectively isotropic.** If light arrives equally from all directions, it does not matter which way each scattering sends it: the scattered light is again the same in all directions, $\oint \mathscr{P} I\, \mathrm{d}\Omega' = J$ for any phase function. So the shape of the phase function (forward or backward scattering) hardly matters deep inside. Its only remaining effect, on the small net flux, is simply to change how effectively scattering blocks the flow (less for forward scattering), as if α^{scat} were replaced by $(1 - g_0)\, \alpha^{\mathrm{scat}}$. We can therefore set $g_0 = 0$.

The net flux in the first approximation

The angle-independent part of the intensity does not contribute to the net flux along z:

$$F_\nu(z) = \int I_\nu^{(1)}(z,\mu)\cos\theta\,\mathrm{d}\Omega = 2\pi\int_{-1}^{+1} I_\nu^{(1)}(z,\mu)\,\mu\,\mathrm{d}\mu$$
$$= -\frac{2\pi}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial z}\int_{-1}^{+1}\mu^2\,\mathrm{d}\mu = -\frac{4\pi}{3\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu(T)}{\partial T}\frac{\partial T}{\partial z} = \frac{4\pi}{3}\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}.$$

Integrating over frequency, the **total flux** is

$$F(z) = \int_0^\infty F_\nu(z)\,\mathrm{d}\nu = -\frac{4\pi}{3}\frac{\partial T}{\partial z}\int_0^\infty \frac{1}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu.$$

Define the **Rosseland mean absorption coefficient**

$$\boxed{\frac{1}{\alpha_R} \equiv \frac{\displaystyle\int_0^\infty \frac{1}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}}$$

The equation of radiative diffusion

Use $\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu = \frac{\partial}{\partial T}\int_0^\infty B_\nu\,\mathrm{d}\nu = \frac{\partial(\sigma T^4/\pi)}{\partial T} = \frac{4\sigma T^3}{\pi}$. Then the Rosseland approximation to the radiative flux is

$$\boxed{F(z) = -\frac{16\sigma T^3}{3\alpha_R}\frac{\partial T}{\partial z} \;\rightarrow\; -\chi\nabla T}$$

the **equation of radiative diffusion**, one of the fundamental equations for stellar interiors.

- It has the form of Fourier's law of heat conduction, with the **radiative conductivity** $\chi = 16\sigma T^3/3\alpha_R$: flux travels *down* the temperature gradient.
- **Per unit mass** (Heng): $\alpha_R = \rho\kappa_\mathrm{R}$. With hydrostatic equilibrium, $\mathrm{d}P = -\rho g\,\mathrm{d}z$,

$$F = \frac{16\,\sigma\,g\,T^3}{3\,\kappa_\mathrm{R}}\frac{\mathrm{d}T}{\mathrm{d}P}.$$

Notation: α is extinction per unit length — Seager's κ, Heng's α_e; $\kappa_\mathrm{R} = \alpha_R/\rho$ is the Rosseland mean per unit mass.

Why a harmonic mean

Three ways to average 1 and 100 (the Pythagorean means):

$$\text{arithmetic } \frac{1+100}{2} = 50.5, \qquad \text{geometric } (1\times 100)^{1/2} = 10, \qquad \text{harmonic } \frac{2}{1+1/100} \approx 2.$$

- The harmonic mean is dominated by the *smallest* value. The Rosseland mean averages $1/\alpha_\nu^{\mathrm{ext}}$, so the frequencies at which the extinction is **small** (transparent) dominate.
- In an optically thick medium radiation leaks out through these **windows**, just as current in parallel resistors flows through the smallest resistance.
- The weighting function $\partial B_\nu/\partial T$ resembles the Planck function but peaks at $h\nu_{\max} = 3.8\,k_{\mathrm{B}}T$ instead of $2.8\,k_{\mathrm{B}}T$.
- Hence a small amount of an opacity source that fills a window (a continuum absorber, a cloud) can change α_R and the deep thermal structure far more than a strong line in an already opaque band.

(2) The Eddington approximation

In the Eddington approximation the intensities are assumed to approach *isotropy*, not necessarily their thermal values (in the Rosseland approximation they approach the Planck function at large depth). Near isotropy can be introduced by taking the intensity linear in μ (frequency suppressed):

$$I(\tau,\mu) = a(\tau) + b(\tau)\,\mu.$$

The first three moments are

$$\text{mean intensity:} \quad J \equiv \tfrac{1}{2}\int_{-1}^{+1} I\,\mathrm{d}\mu = a,$$

$$\text{flux:} \quad H \equiv \tfrac{1}{2}\int_{-1}^{+1} \mu I\,\mathrm{d}\mu = \frac{b}{3},$$

$$\text{radiation pressure:} \quad K \equiv \tfrac{1}{2}\int_{-1}^{+1} \mu^2 I\,\mathrm{d}\mu = \frac{a}{3},$$

so that

$$\boxed{K = \tfrac{1}{3}J} \qquad \text{the } \textbf{Eddington approximation}.$$

Compare the isotropic radiation field, $p = \frac{1}{3}u$, with $p \equiv \frac{1}{c}\int I\cos^2\theta\,\mathrm{d}\Omega$ and $u = \frac{4}{c}I$.

The Eddington approximation: the radiative diffusion equation

The optical depth and the transfer equation are $\mathrm{d}\tau(z) \equiv -\alpha^{\mathrm{ext}}\mathrm{d}z$ and $\mu\,\partial I/\partial\tau = I - S$. The source function does not depend on μ, because $S = (1-\varepsilon)J + \varepsilon B$. Integrating over μ:

$$\tfrac{1}{2}\int_{-1}^{+1}\mathrm{d}\mu\left(\mu\frac{\partial I}{\partial\tau} = I - S\right) \;\rightarrow\; \frac{\partial H}{\partial\tau} = J - S, \qquad \tfrac{1}{2}\int_{-1}^{+1}\mathrm{d}\mu\,\mu\left(\mu\frac{\partial I}{\partial\tau} = I - S\right) \;\rightarrow\; \frac{\partial K}{\partial\tau} = H \;\rightarrow\; \frac{1}{3}\frac{\partial J}{\partial\tau} = H.$$

The two equations combine into

$$\boxed{\frac{1}{3}\frac{\partial^2 J}{\partial\tau^2} = J - S = \varepsilon\,(J - B)}$$

Define a new optical depth

$$\tau_* \equiv \sqrt{3\varepsilon}\,\tau = \sqrt{3\tau_{\mathrm{abs}}(\tau_{\mathrm{abs}} + \tau_{\mathrm{sca}})},$$

and the radiative equation becomes

$$\boxed{\frac{\partial^2 J}{\partial\tau_*^2} = J - B}$$

the **radiative diffusion equation**: given the temperature structure $B(\tau)$ of the medium, it can be solved for J.

The grey atmosphere

If **the opacity is independent of frequency**, illustrative and insightful results can be found; a stellar atmosphere with such an opacity spectrum is a *grey atmosphere*.

- The radiation field still has a frequency distribution, since matter emits a spectrum determined by its temperature.
- We look for the temperature profile (versus depth) and the flux that emanates from a grey atmosphere.
- It is now more convenient to let τ increase *inward* (independent of frequency):

$$\boxed{\mu \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau} = I_\nu - S_\nu}$$

- **(1)** Integrating over all frequencies,

$$\boxed{\mu \frac{\mathrm{d}I}{\mathrm{d}\tau} = I - S} \qquad I = \int_0^\infty I_\nu \,\mathrm{d}\nu, \quad S = \int_0^\infty S_\nu \,\mathrm{d}\nu.$$

Radiative equilibrium

If there are no energy sources or sinks in the atmosphere, the energy absorbed by a layer equals the energy it emits. In the plane-parallel approximation this means the energy generated in the interior passes out as a *constant energy flux* through the outer layers.

$$\iint \alpha_\nu I_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega = \iint j_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega \qquad \text{(energy absorbed or scattered } = \text{ energy radiated)}$$

In the **grey atmosphere**, with the definition of the source function ($j_\nu = \alpha_\nu S_\nu$),

$$\iint \alpha_\nu I_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega = \iint \alpha_\nu S_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega \quad \rightarrow \quad \alpha \int J_\nu \,\mathrm{d}\nu = \alpha \int S_\nu \,\mathrm{d}\nu \quad \Longrightarrow \quad \boxed{J = S}$$

Radiative equilibrium: the moment equations

(2) Average the first equation over the solid angle:

$$\tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu \left(\mu \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau}\right) = \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, I_\nu - \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, S_\nu \;\Rightarrow\; \frac{\mathrm{d}H}{\mathrm{d}\tau} = J - S \xrightarrow{J=S} \boxed{\frac{\mathrm{d}H}{\mathrm{d}\tau} = 0, \quad H = \text{constant}}$$

(3) Multiply the first equation by μ and average over the solid angle:

$$\tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu \left(\mu^2 \frac{\mathrm{d}I}{\mathrm{d}\tau}\right) = \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, (\mu I) - \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, (\mu S) \;\Rightarrow\; \boxed{\frac{\mathrm{d}K}{\mathrm{d}\tau} = H}$$ (isotropic source function).

Integrating gives the radiation pressure as a function of optical depth,

$$\boxed{K(\tau) = H(\tau + c)}$$ (Hc is the constant of integration).

Using the **Eddington approximation**, $K = J/3$, the mean intensity is

$$\boxed{J(\tau) = 3H(\tau + c)}$$ with $c = \frac{2}{3}$, shown on the next slide.

Derivation of the constant of integration

Solve $\mu\,\mathrm{d}I/\mathrm{d}\tau = I - S$ formally. For $\mu > 0$ (outgoing) multiply by $e^{-\tau/\mu}$ and integrate from $\tau' = \infty$ to τ; for $\mu < 0$ (ingoing) integrate from $\tau' = 0$, where nothing comes in:

$$\mu > 0:\ I(\tau,\mu) = \int_{\tau}^{\infty} e^{-(\tau'-\tau)/\mu} S \frac{\mathrm{d}\tau'}{\mu}, \qquad \mu < 0:\ I(\tau,\mu) = \int_{0}^{\tau} e^{-(\tau-\tau')/\mu} S \frac{\mathrm{d}\tau'}{|\mu|}.$$

At the surface, for $\mu > 0$ and with $S = J = 3H(\tau + c)$,

$$I(0,\mu) = \int_0^{\infty} e^{-\tau'/\mu} S \frac{\mathrm{d}\tau'}{\mu} = 3H \int_0^{\infty} e^{-\tau'/\mu} (\tau' + c) \frac{\mathrm{d}\tau'}{\mu} \quad \Longrightarrow \quad \boxed{I(0,\mu) = 3H(\mu + c)}$$

and $I(0,\mu) = 0$ for $\mu < 0$. Now use the definition of the outward flux H at $\tau = 0$:

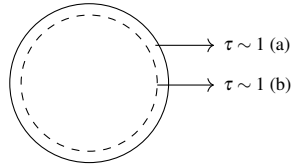
$$H = \tfrac{1}{2}\int_{-1}^{1} \mu I(0,\mu)\,\mathrm{d}\mu = \tfrac{1}{2}\int_{0}^{1} \mu\, I(0,\mu)\,\mathrm{d}\mu = \frac{3H}{2}\left(\frac{1}{3} + \frac{c}{2}\right) \quad \Longrightarrow \quad \boxed{c = \frac{2}{3}}$$

Limb darkening

- Observations show that the intensity at the center of the solar disk is larger than near its limb: **limb darkening**.
- The local temperature increases with depth, and the radiation leaving the star comes from $\tau \approx 1$ along the line of sight. **The layers responsible for the radiation are hotter at the center than near the limb.**
- The Eddington limb-darkening law can be estimated from the grey-atmosphere results (next slide).



Image: Brocken Inaglory.



$T_a > T_b \Rightarrow S_a > S_b$

Limb darkening: the Eddington law

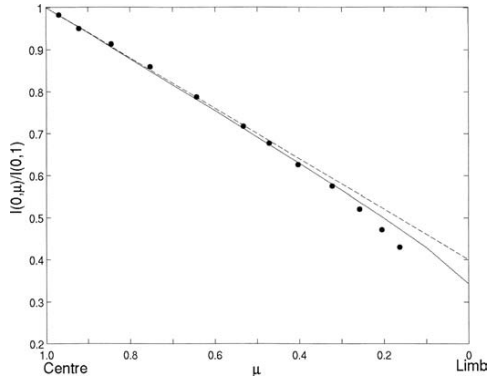
The intensity at the surface of the star is, with $S = J = 3H\,(\tau + \frac{2}{3})$,

$$I(\tau = 0, \mu) = \int_0^\infty S(\tau)\, e^{-\tau/\mu}\, \frac{\mathrm{d}\tau}{\mu}$$
$$= \int_0^\infty 3H\Big(\tau + \frac{2}{3}\Big) e^{-\tau/\mu}\, \mathrm{d}(\tau/\mu) = 3H\Big(\mu + \frac{2}{3}\Big).$$

The direction toward the center of the disk is $\theta = 0$, $\mu = 1$; the limb corresponds to $\mu < 1$. The ratio of the intensities is

$$\boxed{\frac{I(0,\mu)}{I(0,1)} = \frac{3}{5}\Big(\mu + \frac{2}{3}\Big)}$$

the **Eddington limb-darkening law**. At the limb ($\mu = 0$) the ratio is 0.4: the intensity of the grey-atmosphere approximation at the limb is 40% of its value at the center of the disk.



Dots: observation. Dashed: Eddington approximation. Solid: exact grey-atmosphere solution; $\lambda = 5485$ Å (Pierce et al. 1950).

The temperature profile of a grey atmosphere

From the definition of the effective temperature,

$$H = \frac{1}{4\pi}\int \cos\theta\, I\, \mathrm{d}\Omega = \frac{1}{4\pi}F = \frac{1}{4\pi}\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4.$$

The relation between the intensity and the outward flux, $F = \pi J$, gives

$$J(\tau) = \frac{1}{\pi}F(\tau) = \frac{1}{\pi}\sigma_{\mathrm{SB}} T^4(\tau),$$

where T is the effective temperature at τ. Combining with $J(\tau) = 3H\left(\tau + \frac{2}{3}\right)$,

$$T^4 = \frac{3}{4}T_{\mathrm{eff}}^4\left(\tau + \frac{2}{3}\right) \quad \Longrightarrow \quad \boxed{T = T_{\mathrm{eff}}\left[\frac{3}{4}\left(\tau + \frac{2}{3}\right)\right]^{1/4}}$$

This tells us how the temperature varies inside a grey atmosphere.

- At $\tau = 2/3$, $T = T_{\mathrm{eff}}$: the temperature at the photosphere is the effective temperature.
- The same form follows from radiative diffusion alone: **Milne's solution** (next slide).

Milne's solution

Deep in the atmosphere (no external flux), the flux is carried outward by radiation and, in radiative equilibrium, is constant with depth; set it to the internal heat flux, $F = \sigma_{\mathrm{SB}} T_{\mathrm{int}}^4$. With the Rosseland optical depth $\mathrm{d}\tau_R = -\alpha_R \,\mathrm{d}z$, the diffusion equation reads

$$\sigma_{\mathrm{SB}} T_{\mathrm{int}}^4 = \frac{16\sigma_{\mathrm{SB}} T^3}{3} \frac{\mathrm{d}T}{\mathrm{d}\tau_R} = \frac{4\sigma_{\mathrm{SB}}}{3} \frac{\mathrm{d}T^4}{\mathrm{d}\tau_R},$$

and integrating over τ_R,

$$\boxed{T^4 = \frac{3}{4} T_{\mathrm{int}}^4 \,(\tau_R + C)}$$

with $C \sim 1$ a constant of integration.

- This is **Milne's solution**, first derived for stars. Diffusion alone cannot determine C, because it does not hold near the surface. The Eddington approximation with the boundary condition of no incoming radiation gives $C = 2/3$: exactly the grey-atmosphere profile of the previous slide, with $T_{\mathrm{int}} = T_{\mathrm{eff}}$.
- At $\tau_R \approx 1$ the observer measures $T \approx T_{\mathrm{int}}$: for the Sun, $T_{\mathrm{int}} \approx 5800\,\mathrm{K}$ is the photospheric temperature. Temperature rises as $\tau_R^{1/4}$ with depth.
- In an irradiated planet this internal part is joined by the absorbed starlight; combining the two gives the temperature–pressure profiles later in this lecture.

Boundary conditions: the two-stream approximation

To solve the second-order equation for J we need two boundary conditions. One way is the **two-stream approximation**: the whole radiation field is represented by radiation at just two angles, $\mu = \pm\mu_0$,

$$\boxed{I(\tau,\mu) = I^+(\tau)\,\delta(\mu-\mu_0) + I^-(\tau)\,\delta(\mu+\mu_0)}$$

The two terms are the outward and inward intensities. The three moments are

$$J = \tfrac{1}{2}\left(I^+ + I^-\right), \qquad H = \tfrac{1}{2}\mu_0\left(I^+ - I^-\right), \qquad K = \tfrac{1}{2}\mu_0^2\left(I^+ + I^-\right),$$

and $K = \frac{1}{3}J$ requires $\mu_0 = 1/\sqrt{3}$ ($\theta_0 = \cos^{-1}\mu_0 = 54.74°$).

1. Solve the first two for $I^\pm$: $I^+ + I^- = 2J$ and $I^+ - I^- = 2H/\mu_0$, so $I^\pm = J \pm H/\mu_0$.
2. The first moment of $\mu\,\partial I/\partial\tau = I - S$ (multiply by μ and average over angle) gives $\partial K/\partial\tau = H$; with $K = \frac{1}{3}J$, $H = \frac{1}{3}\,\partial J/\partial\tau$.
3. Hence $H/\mu_0 = \sqrt{3}\cdot\frac{1}{3}\,\partial J/\partial\tau = \frac{1}{\sqrt{3}}\,\partial J/\partial\tau$, and

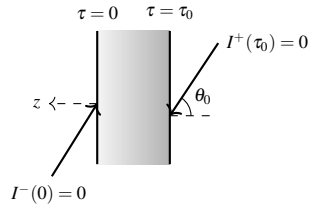
$$\boxed{I^\pm = J \pm \frac{1}{\sqrt{3}}\frac{\partial J}{\partial\tau}}$$

Boundary conditions: a slab with no incident radiation

Let the medium extend from $\tau = 0$ to $\tau = \tau_0$ with no incident radiation, $I^-(0) = 0$ and $I^+(\tau_0) = 0$; with $I^{\pm} = J \pm \frac{1}{\sqrt{3}} \partial J / \partial \tau$ this gives

$$\boxed{\begin{aligned} \frac{1}{\sqrt{3}} \frac{\partial J}{\partial \tau} &= J \quad \text{at } \tau = 0, \\ \frac{1}{\sqrt{3}} \frac{\partial J}{\partial \tau} &= -J \quad \text{at } \tau = \tau_0. \end{aligned}}$$

Note the generalization to N streams, in which the field is represented at N angles.



The temperature–pressure (T–P) profile

The vertical temperature structure of an atmosphere is fundamental for several reasons.

- **Habitability.** The surface temperature and pressure decide whether liquid water is stable, and whether complex molecules can survive at all.
- **Chemistry.** The T–P structure is the starting point for equilibrium and non-equilibrium chemistry, i.e. for the composition.
- **Spectra.** As Lectures 5 and 6 showed, the emergent spectrum samples the temperature at the depth where each wavelength becomes optically thick; absorption or emission features depend on the *temperature gradient*.

Pressure is used as a proxy for altitude (the two are linked by hydrostatic equilibrium), hence the abbreviation **T–P profile**. There is no single equation for it: radiation, convection, gravity and the opacities all play a part.

This lecture treats the **one-dimensional** vertical structure. Hot, tidally locked exoplanets also have strong *horizontal* contrasts (permanent day and night sides); horizontal heat transport is the subject of atmospheric circulation.

Solar System profiles look alike

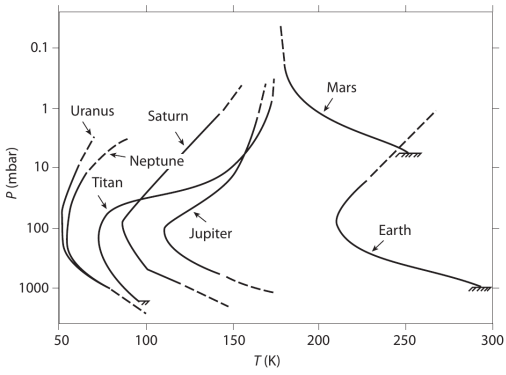


Figure 1: Vertical temperature structure of Solar System planet atmospheres. Seager (2010), Fig. 9.1, after Goody & Yung.

- Every profile has a **troposphere** where temperature falls with height, a **tropopause** minimum near 0.1 bar, and in most cases a **stratosphere** where it rises again.
- The thermal inversion comes from absorption of stellar ultraviolet: by **CH_4-induced hazes** in the giant planets and Titan, by **O_3** on Earth. Mars, with neither, has no stratospheric inversion.
- The similarity is not a coincidence: the same physics — hydrostatic equilibrium, radiative transfer, convection — shapes all of them (Seager's exercise 8). The aim of this lecture is to understand that physics.

Earth's vertical structure

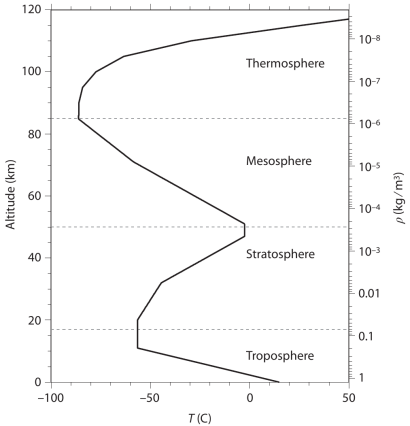


Figure 2: Earth's atmosphere at mid-latitudes (1976 U.S. Standard Atmosphere). Seager (2010), Fig. 9.2.

The layers are named by where the temperature gradient reverses; the reversals come from *where* solar radiation is absorbed, which depends on wavelength and composition.

- **Troposphere** (0–∼12 km): where we live and weather happens; holds ∼ 85% of the mass, because density falls exponentially with height. Most strong visible and infrared spectral features form here. Mt. Everest (8.8 km) does not reach the top.
- Temperature is highest at the ground: visible sunlight reaches the surface, which re-radiates in the infrared and heats the air from below (plus the greenhouse effect).
- **Stratosphere**: temperature *increases* with height, because O_3 absorbs solar UV. This also shields life from UV.
- Mesosphere and thermosphere above matter mainly through photochemistry and atmospheric escape.

Where sunlight is absorbed

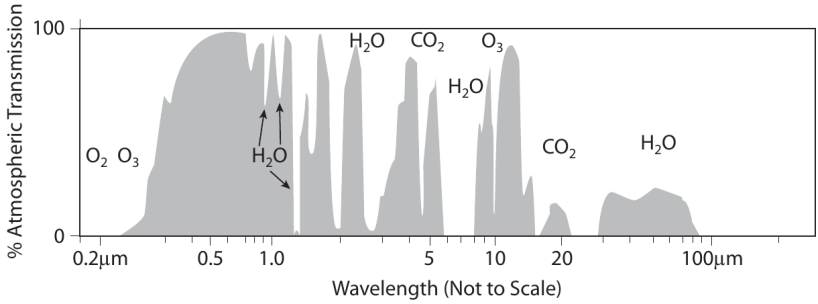


Figure 3: Transmission of solar radiation through Earth's atmosphere, with the main absorbers. Seager (2010), Fig. 9.3.

- The visible window (~ 0.3–$1\,\mu$m) is nearly transparent: most sunlight reaches and heats the **ground**.
- O_2 and O_3 absorb the ultraviolet **high up**, heating the stratosphere and thermosphere.
- In the infrared, H_2O and CO_2 make the atmosphere largely opaque, except in a few windows (e.g. 8–12 μm): the surface emission is absorbed and re-emitted by the air.

Hydrostatic equilibrium

Gravity compresses the atmosphere; the gas pressure resists. Without the pressure force the atmosphere would collapse into a thin shell, without gravity it would diffuse into space. “Static” means no vertical acceleration.

For a slab of area δA and thickness δz (height z upward), the pressure difference between its faces, $\mathrm{d}P = (\mathrm{d}P/\mathrm{d}z)\,\delta z$, must hold up its weight $\rho g\,\delta z\,\delta A$:

$$\frac{\mathrm{d}P}{\mathrm{d}z}\,\delta z\,\delta A = -g\rho\,\delta z\,\delta A \qquad \Longrightarrow \qquad \boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -g\rho}$$

In terms of the optical depth, $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$ with κ the (mean) extinction per unit length, or per unit mass $\kappa_m = \kappa/\rho$:

$$\frac{\mathrm{d}P}{\mathrm{d}\tau} = \frac{g\rho}{\kappa} = \frac{g}{\kappa_m} \qquad \Longrightarrow \qquad \tau = \frac{\kappa_m P}{g} \text{ for constant } \kappa_m.$$

- Valid almost everywhere in a planetary atmosphere. Exceptions have real vertical accelerations: hydrodynamic escape in the upper atmosphere, and small-scale vigorous systems (convective plumes, thunderstorms, tornadoes).

Equation of state and the pressure scale height

For planetary atmospheres the ideal gas law is adequate,

$$P = nkT = \frac{\rho kT}{\mu_m m_{\rm H}}, \qquad \mu_m \equiv \frac{\bar{m}}{m_{\rm H}} \quad \text{(mean molecular weight)}.$$

Combining with hydrostatic equilibrium,

$$\frac{{\rm d}P}{P} = -\frac{\mu_m m_{\rm H} g}{kT}\,{\rm d}z \quad \Longrightarrow \quad \boxed{P = P_0\,{\rm e}^{-z/H}, \qquad H \equiv \frac{kT}{\mu_m m_{\rm H} g}}$$

for constant T, μ_m and g. H is the height over which pressure (and density) fall by a factor e.

- Temperature varies by only a factor of a few through an atmosphere, so P and ρ still fall roughly exponentially.
- The observable spectrum comes from a region a few scale heights thick (roughly $5H$).

How large is H? (Seager, exercise 2)

planet	T (K)	g (m s^{-2})	μ_m	H (km)
Earth (N_2/O_2)	288	9.8	29	8.4
hot Jupiter (H_2/He)	1500	25	2.3	220
hot super-Earth, H_2-rich	1000	20	2.3	180
hot super-Earth, CO_2-rich	1000	20	44	9.4

- $H \propto T/(\mu_m g)$: hot, low-gravity planets with light (hydrogen) atmospheres are the puffiest.
- The **mean molecular weight** matters as much as the temperature: the same hot super-Earth has a 20 times larger H if its atmosphere is H_2-rich. Since transmission features scale as H (Lecture 1), this is how transit spectra distinguish hydrogen-rich from heavy atmospheres.
- Solar System values are 5–20 km; hot Jupiters reach hundreds of kilometres.

Emission temperature

Recall the energy balance of Lecture 4: with uniform redistribution of the absorbed starlight,

$$\sigma T_{\rm eq}^4 = \frac{1}{4}\,\sigma T_*^4 \left(\frac{R_*}{a}\right)^2 (1-A_{\rm B}) \qquad \Longrightarrow \qquad T_{\rm eq} = T_* \left(\frac{R_*}{2a}\right)^{1/2} (1-A_{\rm B})^{1/4}.$$

- $T_{\rm eq}$ is the temperature of an isothermal planet in complete equilibrium with its star. In an idealized atmosphere it is the temperature of the layer from which most of the radiation escapes to space: the **emission temperature** $T_e \equiv T_{\rm eq}$.
- Question: how much warmer than T_e is the *surface*? We answer it with one-layer models (after Marshall & Plumb), assuming
  - the star emits mostly in the visible, and all unscattered starlight reaches and heats the ground;
  - the atmosphere absorbs and emits only in the infrared.

The simple greenhouse

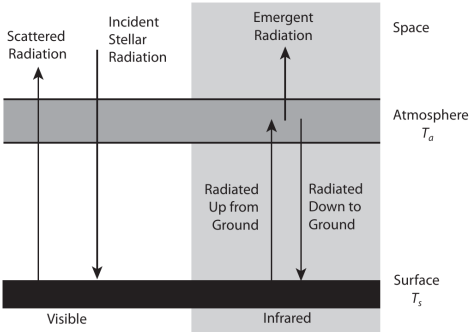


Figure 4: The simple greenhouse: one infrared-opaque layer at T_a above a surface at T_s. Seager (2010), Fig. 9.4.

The atmosphere layer is transparent in the visible and opaque (a black body) in the infrared.

- **Top of the atmosphere:** the layer is what space sees, so it must emit the absorbed starlight: $\sigma T_a^4 = \sigma T_e^4$, i.e. $T_a = T_e$.
- **Surface:** it absorbs the starlight *and* the downward emission of the layer:

$$\sigma T_s^4 = \sigma T_e^4 + \sigma T_a^4 = 2\sigma T_e^4.$$

$$\boxed{T_s = 2^{1/4} T_e \approx 1.19\, T_e}$$

- Earth: $T_e \simeq 255\,\mathrm{K}$ gives $T_s \simeq 303\,\mathrm{K}$ (observed $\simeq 288\,\mathrm{K}$): too warm.
- Venus: $T_e \simeq 230\,\mathrm{K}$ gives $T_s \simeq 274\,\mathrm{K}$ (observed $\simeq 730\,\mathrm{K}$): far too cold.

The leaky greenhouse

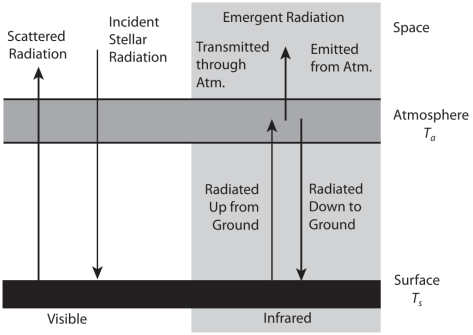


Figure 5: The leaky greenhouse: the layer absorbs only a fraction α of the infrared. Seager (2010), Fig. 9.5.

Now the layer absorbs a fraction $\alpha \in (0,1)$ of the infrared and, by Kirchhoff's law, emits with the same emissivity α.

$$\text{top:} \quad \sigma T_e^4 = \alpha \sigma T_a^4 + (1-\alpha)\,\sigma T_s^4,$$
$$\text{surface:} \quad \sigma T_s^4 = \sigma T_e^4 + \alpha \sigma T_a^4.$$

Subtracting eliminates T_a:

$$\boxed{T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e}, \qquad T_a = \left(\frac{1}{2-\alpha}\right)^{1/4} T_e.$$

- $\alpha \to 1$: the simple greenhouse; $\alpha \to 0$: no atmosphere, $T_s = T_e$. Always $T_s \geq T_e \geq T_a$.

Seager's Eqs. (9.20)–(9.21) omit the emissivity α on the σT_a^4 terms. T_s comes out the same, but her implied T_a ($T_a^4 = \alpha T_e^4/(2-\alpha)$) does not satisfy $T_a \leq T_e$ in general.

Many layers: towards radiative equilibrium

Stack n infrared-opaque layers above the surface (Seager's exercise 3). Each layer receives from its two neighbours and emits up and down; the top layer emits σT_e^4. Working down, the k-th layer from the top has $T_k^4 = k\,T_e^4$, and the surface

$$\boxed{T_s = (n+1)^{1/4}\,T_e}$$

- Venus: $(T_s/T_e)^4 = (730/230)^4 \approx 100$, i.e. the equivalent of ~ 100 opaque layers — an infrared optical depth of order 100 above the surface. The massive CO_2 atmosphere (92 bar) is that opaque.
- With $n \to$ many layers of appropriate absorptivity, at all wavelengths, with molecules and clouds, these models become a **multilayer radiative-equilibrium model**: the subject of the next sections.
- The one-layer result also reappears in the continuous grey model: its top temperature is $2^{-1/4}$ times the bottom temperature.

Which process carries the energy?

Energy can be transported by radiation, convection, or conduction; conduction is negligible in the dense lower atmosphere. Energy takes the **path of least resistance**: if the opacity is low, photons carry it; if the opacity is high, radiation is impeded and convection takes over.

Stability of a displaced parcel. Lift a parcel slightly: it expands *adiabatically* to the ambient pressure, keeping its own temperature and density.

- Colder (denser) than its new surroundings: it sinks back. The atmosphere is **stable** against convection.
- Warmer (lighter) than its surroundings: buoyancy keeps it rising. The atmosphere is **unstable**; convection occurs.

The parcel cools at the adiabatic rate, so the atmosphere is unstable when its own temperature falls with height *faster* than the adiabat:

$$\left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} > \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}}$$

(Schwarzschild criterion).

Stable and unstable profiles

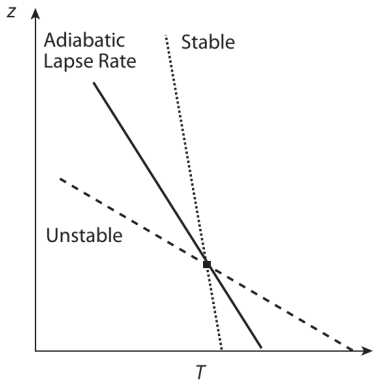


Figure 6: Profiles stable (dotted) or unstable (dashed) against convection, relative to the adiabat. Seager (2010), Fig. 9.6.

The two gradients to compare:

$$\left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{ad}} = -\frac{g}{c_p}, \qquad \left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{rad}} = -\frac{3}{16}\frac{\kappa F_{\mathrm{int}}}{\sigma T^3},$$

the second from radiative diffusion, valid deep in the atmosphere. Convection sets in when

- the **adiabatic gradient is small**: large c_p, e.g. where hydrogen ionizes and absorbed heat goes into ionization rather than temperature (stars; giant-planet interiors above $\sim 4000\,\mathrm{K}$);
- the **radiative gradient is large**: large flux F or large opacity κ.

Seager gives the units of c_p as $\mathrm{J\,kg^{-1}\,s^{-1}}$; they are $\mathrm{J\,kg^{-1}\,K^{-1}}$.

Convection in planets

- **Opacity drives convection.** Opacity grows with the density of absorbers, so the deep, optically thick layers are almost always convective, while radiation dominates higher up. Opacity can also change with T and P as the composition changes.
- **Jupiter:** radiative upper atmosphere over a convective interior — but perhaps another radiative zone underneath, where the mean opacity has a minimum ($T \approx 1300$–$1800\,\mathrm{K}$; Guillot et al. 1994).
- **Hot Jupiters:** strong irradiation makes the dayside profile nearly isothermal or even inverted, so the radiative zone extends *deep* into the atmosphere.
- **Rocky planets with thin atmospheres:** most starlight is absorbed at the surface, which becomes much hotter than the air above. The large upward flux from the ground makes the radiative gradient steep, and a **convective layer** forms just above the surface — on Earth most of the troposphere.
- **Inversions are very stable:** a lifted parcel cools as it expands while its surroundings get warmer with height, so it sinks back strongly. Stratospheres do not convect.

Radiative equilibrium

If energy is carried only by radiation, no energy is created or destroyed in a layer: what enters a volume must leave it. Equivalently, the net flux is constant,

$$\frac{\mathrm{d}F(\tau)}{\mathrm{d}\tau} = 0.$$

Total absorption per volume is $\int\!\!\int \kappa_\nu I_\nu \,\mathrm{d}\Omega\,\mathrm{d}\nu = 4\pi \int \kappa_\nu J_\nu \,\mathrm{d}\nu$; with isotropic emission and $\varepsilon_\nu = \kappa_\nu S_\nu$, total emission is $4\pi \int \kappa_\nu S_\nu \,\mathrm{d}\nu$. Hence

$$\boxed{\int_0^\infty \kappa_\nu \left(J_\nu - S_\nu\right) \mathrm{d}\nu = 0}$$

- Absorption and emission balance *in total*, not at each frequency: a layer can absorb in the visible and emit in the infrared.
- The constraint involves J_ν, so it must be solved *together* with the transfer equation. The opacity's frequency dependence enters directly, and an inversion can arise naturally from the solution.
- The thermal (infrared) flux at the top is $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$: the interior heat plus the reradiated starlight. Interior heat comes from slow loss of formation energy (Jupiter emits about 1.7 times what it absorbs) or radioactivity (Earth); for Earth and close-in planets the stellar term dominates.

No general analytic solution exists; the **grey** atmosphere gives one.

Grey atmosphere heated from below

Assume LTE, no scattering, and frequency-independent opacity (grey). The energy comes from below only: from the interior of a giant planet ($T_{\rm int} = T_{\rm eff}$), or from a ground heated by sunlight that the atmosphere does not absorb ($T_{\rm int} = T_s$). Radiative equilibrium gives

$$J(\tau) = S(\tau) = B(T) = \frac{\sigma T^4}{\pi}.$$

This is exactly the grey atmosphere derived above: the moments of $\mu\,\mathrm{d}I/\mathrm{d}\tau = I - J$ give a constant flux $F = \sigma T_{\rm int}^4$ and $\mathrm{d}K/\mathrm{d}\tau = F/4\pi$; with $K = J/3$ and the surface boundary condition,

$$\boxed{T^4 = \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3}\right)}$$

- **Top of the atmosphere:** $T(0)^4 = T_{\rm int}^4/2$, the same $2^{-1/4}$ ratio as the simple greenhouse.
- **Gradient:** differentiating with $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$, $\mathrm{d}T/\mathrm{d}z = -\frac{3}{16}\kappa T_{\rm int}^4/T^3$ — the radiative-diffusion gradient. Large flux or large opacity gives a steep gradient.
- With the opacity's T, P dependence and hydrostatic equilibrium, $T(\tau)$ becomes a T–P profile.

Why T_{eq} is not enough: a myriad of effects

Without data, an exoplanet is labelled by its equilibrium temperature, computable without knowing anything about its atmosphere. The Solar System shows how misleading that can be: Earth's T_{eq} is below freezing, yet it has liquid water; Venus's T_{eq} is barely higher, yet its surface is about 700 K. Real atmospheres contain many opacity sources that warm or cool them (Heng, Ch. 4):

- **Greenhouse warming:** gases such as H_2O, CO_2 and CH_4 absorb the planet's *thermal* emission.
- **Anti-greenhouse cooling:** enhanced absorption, or scattering, of *starlight* high in the atmosphere.
- **Scattering greenhouse:** thermal emission is scattered back toward the planet.
- **Aerosols and clouds** combine all of these, and can cool or heat depending on their properties and where they sit.

The aim, following Pierrehumbert, is to understand this richness with simple, clear models that isolate one effect at a time — before, or instead of, a full simulation.

Why two bands? The dual-band approximation

Analytic profiles rest on the assumption that starlight and the planet's thermal emission occupy *distinctly different* wavelengths. Wien's law supports it:

$$\lambda_{\mathrm{peak}} \approx 0.5\,\mu\mathrm{m}\left(\frac{T}{6000\,\mathrm{K}}\right)^{-1} \qquad (\lambda_{\mathrm{peak}} T = 2898\,\mu\mathrm{m\,K}).$$

- Sun-like star: $\sim 0.5\,\mu$m. Earth (~ 300 K): $\sim 10\,\mu$m. Hot exoplanet (~ 1500 K): $\sim 2\,\mu$m.
- Heng calls the stellar range the **shortwave** (subscript S) and the planetary range the **longwave** (subscript L). They are usually the visible and the infrared — Seager's "vis" and "ir" bands below.
- **Caveat:** an M dwarf ($\lesssim 3000$ K) peaks in the near-infrared, so both bands can lie in the infrared; the separation is then less clean.

Within each band the opacity is replaced by a single mean value (a "double-gray" atmosphere), and each band can have its own scattering properties.

Heated from above and below: two bands

For an irradiated planet the starlight is absorbed *within* the atmosphere. The star is hotter than the planet, so starlight is absorbed at short wavelengths and re-emitted in the infrared. Treat two grey bands (Hansen 2008):

- **visible** ($I_{\rm vis}$, opacity $\kappa_{\rm vis}$): absorbed only, never emitted, $S_{\rm vis} = 0$;
- **infrared** ($I_{\rm ir}$, opacity $\kappa_{\rm ir}$): absorbed and re-emitted.

Radiative equilibrium now balances visible plus infrared absorption against infrared emission:

$$\kappa_{\rm vis} J_{\rm vis} + \kappa_{\rm ir} J_{\rm ir} = \kappa_{\rm ir} S_{\rm ir} \quad \Longrightarrow \quad \boxed{S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}, \qquad \gamma \equiv \frac{\kappa_{\rm vis}}{\kappa_{\rm ir}}}$$

With $\tau \equiv \tau_{\rm ir}$ and $\mathrm{d}\tau_{\rm vis} = \gamma \mathrm{d}\tau$, the stellar beam from direction μ_0 is only attenuated:

$$J_{\rm vis} = \frac{I_0}{4\pi} \mathrm{e}^{-\gamma\tau/\mu_0}, \qquad F_{\rm vis} = -\mu_0 I_0 \mathrm{e}^{-\gamma\tau/\mu_0} \quad \text{(downward)}.$$

Heated from above and below: the solution

Zeroth moment (IR): $\frac{1}{4\pi}\,\mathrm{d}F_{\mathrm{ir}}/\mathrm{d}\tau = J_{\mathrm{ir}} - S_{\mathrm{ir}} = -\gamma J_{\mathrm{vis}}$, so

$$F_{\mathrm{ir}}(\tau) = \mu_0 I_0\,\mathrm{e}^{-\gamma\tau/\mu_0} + F_{\mathrm{int}}, \qquad F_{\mathrm{ir}} + F_{\mathrm{vis}} = F_{\mathrm{int}} = \sigma T_{\mathrm{int}}^4 = \text{const.}$$

First moment with $K = J/3$: $\mathrm{d}J_{\mathrm{ir}}/\mathrm{d}\tau = (3/4\pi)F_{\mathrm{ir}}$,

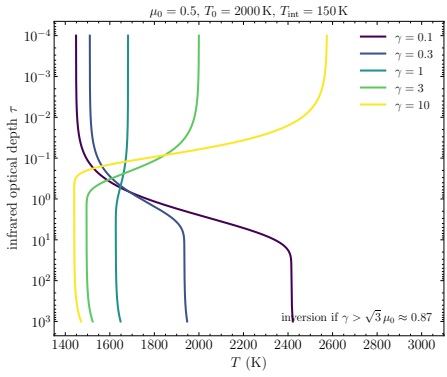
$$J_{\mathrm{ir}}(\tau) = \frac{3}{4\pi}\left[F_{\mathrm{int}}\,\tau - \frac{\mu_0^2}{\gamma} I_0\,\mathrm{e}^{-\gamma\tau/\mu_0}\right] + C.$$

Then $\sigma T^4/\pi = S_{\mathrm{ir}} = J_{\mathrm{ir}} + \gamma J_{\mathrm{vis}}$. Fixing C with the Eddington boundary condition $J_{\mathrm{ir}}(0) = F_{\mathrm{ir}}(0)/2\pi$ and writing $I_0 = \sigma T_0^4$,

$$\boxed{T^4 = \frac{3}{4} T_{\mathrm{int}}^4 \left(\tau + \frac{2}{3}\right) + \frac{\mu_0 T_0^4}{4}\left[2 + \frac{3\mu_0}{\gamma} + \left(\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma}\right)\mathrm{e}^{-\gamma\tau/\mu_0}\right]}$$

Seager's Eq. (9.60) sets $\sigma T^4/\pi = J_{\mathrm{ir}}$, dropping the γJ_{vis} term required by her own Eq. (9.48). Her Eq. (9.62), after Hansen (2008), fixes C from the exact emergent flux rather than the Eddington boundary condition, which adds terms in $\ln(1+\gamma/\mu_0)$; the behaviour is the same.

What the irradiated profile shows



The boxed solution for several γ. Script

`scripts/make_irradiated_grey_profiles.py`.

- **Deep layers** ($\gamma\tau/\mu_0 \gg 1$): the starlight has been absorbed; $T^4 \to \frac{3}{4} T_{\mathrm{int}}^4 \tau + \mathrm{const}$ — the interior-heated profile, nearly isothermal when T_{int} is small.
- **Visible transparent** ($\gamma \to 0$): the profile tends to $\frac{3}{4}(T_{\mathrm{int}}^4 + \mu_0 T_0^4)(\tau + \frac{2}{3})$, as if all the flux came from below — the ground-heated case.
- **Thermal inversion** at the top when

$$\gamma > \sqrt{3}\,\mu_0,$$

  i.e. when starlight is absorbed higher up than the infrared can escape (strong optical absorbers: TiO/VO, metals, hazes).
- Smaller μ_0 (grazing starlight, near the terminator) deposits the energy higher and lowers the deep temperature.

Irradiated grey T–P profiles

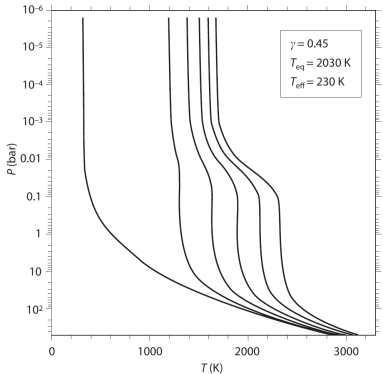


Figure 7: Grey-atmosphere T–P profiles for $\mu_0 = 0.0$–1.0 (left to right), $\gamma = 0.45$. Seager (2010), Fig. 9.7, after Hansen (2008).

- Converting τ to P with hydrostatic equilibrium, $\tau \simeq \kappa_m P/g$, turns $T(\tau)$ into a T–P profile.
- With $\gamma = 0.45 < \sqrt{3}\,\mu_0$ (for most μ_0) there is no inversion: an isothermal upper atmosphere, a transition where the starlight is absorbed ($\gamma\tau/\mu_0 \sim 1$), and a deep interior-heated part.
- A larger μ_0 (starlight closer to vertical) penetrates deeper and gives a hotter profile.
- Such analytic profiles are the usual **starting guess** for full radiative-convective models, and are still widely used for interpreting exoplanet spectra.

Scattering in a band: one parameter, and the Bond albedo

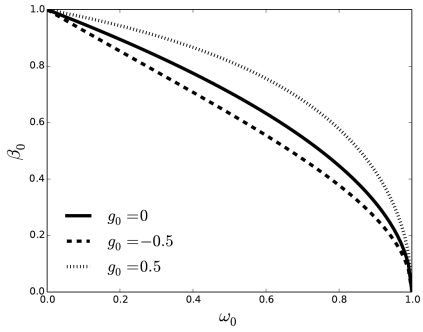


Figure 8: The scattering parameter versus ω_0 for isotropic ($g_0 = 0$), forward and backward scattering. Heng (2017), Fig. 4.1.

In the two-stream equations the single-scattering albedo ω_0 and the asymmetry factor g_0 never appear separately, only as the **scattering parameter**

$$\beta_0 \equiv \left(\frac{1-\omega_0}{1-\omega_0 g_0} \right)^{1/2},$$

with $\beta_0 = 1$ for pure absorption and 0 for pure scattering.

For starlight falling on an opaque atmosphere the two-stream solution gives the reflected fraction; averaged over the stellar spectrum,

$$\boxed{A_B = \frac{1-\beta_{S0}}{1+\beta_{S0}}}$$

- With $g_0 = 0$ this is the reflectivity ζ_-/ζ_+ of a thick slab in the two-stream test case (supplementary notes): 0.82, 0.52, 0.17 for $\omega_0 = 0.99, 0.9, 0.5$.
- Forward scattering ($g_0 > 0$) raises β_0: a cloud of forward-scattering particles is darker than its ω_0 alone suggests.

Where is the starlight absorbed? The photon deposition depth

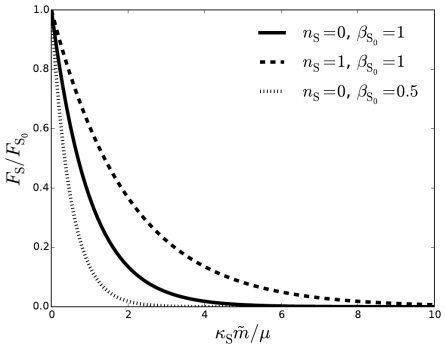


Figure 9: Generalized Beer's law. Scattering ($\beta_{S0} < 1$) makes the starlight fade faster; an opacity that falls with height ($n_S = 1$) lets it reach deeper. Heng (2017), Fig. 4.2.

Use the column mass $\tilde{m} = P/g$ as the depth coordinate. For a constant shortwave opacity κ_S (per unit mass) the incident stellar flux decays as a **generalized Beer's law**,

$$F_S = F_{S0} \exp\Big(- \frac{\kappa_S \tilde{m}}{|\mu|\,\beta_{S0}} \Big), \qquad F_{S0} = |\mu|\,\sigma T_{\rm irr}^4,$$

where $T_{\rm irr} = \sqrt{2}\,T_{\rm eq}$ is the substellar **irradiation temperature**. Scattering ($\beta_{S0} < 1$) shortens the path.

Averaged over the dayside, the flux falls as $2E_3(\kappa_S\tilde{m}/\beta_{S0})$, by $1/\mathrm{e}$ at $\kappa_S\tilde{m}/\beta_{S0} \approx 0.63$: the **photon deposition depth**

$$P_D \approx \frac{0.63\,g\,\beta_{S0}}{\kappa_S} = \frac{0.63\,g}{\kappa_S}\,\frac{1-A_{\rm B}}{1+A_{\rm B}}$$

Stronger scattering (higher albedo), or backward scattering, deposits the starlight *higher* (lower P).

An analytic model for controlled experiments

Heng derives the global-mean T–P profile of a dual-band atmosphere with scattering in both bands (Heng et al. 2014). Its inputs are few:

- $T_{\rm irr}$ (irradiation), $T_{\rm int}$ (interior heat), g;
- the shortwave opacity κ_S and the longwave opacity

$$\kappa_L = \kappa_0 + \kappa_{\rm CIA}\frac{\tilde{m}}{\tilde{m}_0},$$

  a constant part for greenhouse gases (H_2O, CH_4, CO) plus a part growing with pressure for collision-induced absorption;
- the scattering parameters β_{S0} and β_{L0}.

The solution is a sum of terms: an interior term, which is **Milne's solution** generalized to scattering, and stellar-heating terms containing E_2, E_3, E_4 of $\kappa_S\tilde{m}/\beta_{S0}$.

Fiducial model: $T_{\rm int} = 0$, $T_{\rm irr} = 1200\,{\rm K}$, $g = 10^3\,{\rm cm\,s^{-2}}$, $\kappa_S = 0.01$, $\kappa_0 = 0.02\,{\rm cm^2\,g^{-1}}$, no CIA, no scattering. Changing *one* parameter at a time isolates its effect, like a laboratory experiment.

The model sets the absorption, flux and Planck mean opacities equal; recovering Milne's solution, which needs the Rosseland mean, then implies these arithmetic means equal a harmonic one — the price of an analytic model.

Greenhouse warming and collision-induced absorption

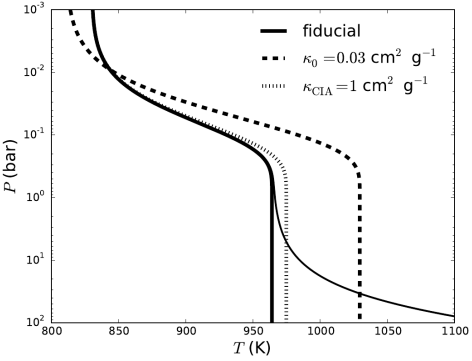


Figure 10: Fiducial model (thick solid), larger longwave opacity (dashed), CIA switched on (dotted); thin solid: fiducial with $T_{\rm int} = 150$ K. Heng (2017), Fig. 4.3.

- **Generic shape:** T rises with pressure, then turns *isothermal* below the depth where starlight is used up. With interior heat a deep "tail" appears: Milne's solution.
- Without an atmosphere the planet would be at $T_{\rm irr}/\sqrt{2} \approx 850$ K. The deep atmosphere is hotter: starlight enters easily, thermal emission escapes with difficulty — the **greenhouse effect**.
- **More longwave opacity** (dashed): stronger greenhouse; the lower atmosphere warms and the upper atmosphere cools.
- **Collision-induced absorption** (dotted): H_2 or N_2 have no permanent dipole, but in collisions at $\gtrsim 1$ bar a transient pair absorbs. A two-body process, so $\tau_{\rm CIA} \propto P^2$; it is a broad continuum and warms the deep layers most. First seen in Uranus and Neptune.

Anti-greenhouse cooling and thermal inversions

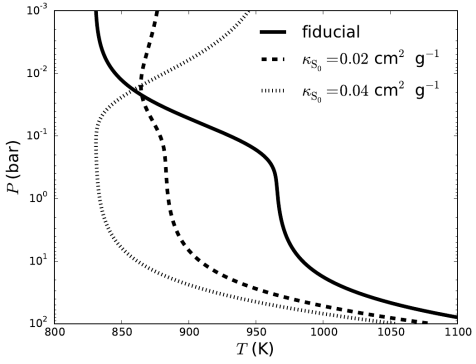


Figure 11: Increasing the shortwave opacity: $\kappa_S = \kappa_0$ (dashed) and $\kappa_S = 2\kappa_0$ (dotted), with $T_{\rm int} = 150$ K. Heng (2017), Fig. 4.4.

- Add absorbers of *starlight* until the shortwave opacity exceeds the longwave one: the **anti-greenhouse effect**. The upper atmosphere warms, the lower atmosphere cools.
- The isothermal deep part starts at lower pressure, because starlight is deposited higher up ($P_D \propto 1/\kappa_S$).
- For large enough κ_S/κ_L the profile develops a **thermal inversion** — the same physics as the criterion $\gamma > \sqrt{3}\,\mu_0$ found earlier.
- Earth: O_3 absorbing solar UV. Hot Jupiters: TiO/VO, by analogy with brown dwarfs, and atomic metals; the absorber is still debated.
- A warm upper atmosphere also shortens chemical timescales, which can undo disequilibrium chemistry — inversions matter for interpreting abundances.

Albedo and the scattering greenhouse

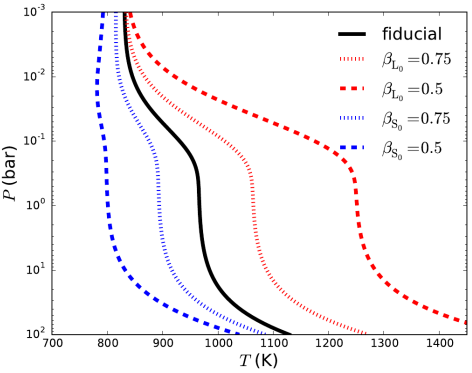


Figure 12: Shortwave scattering (blue, $\beta_{S0} = 0.75, 0.5$) and longwave scattering (red, $\beta_{L0} = 0.75, 0.5$), with $T_{\rm int} = 150\,{\rm K}$. Heng (2017), Fig. 4.5.

- **Shortwave scattering** raises the Bond albedo: $\beta_{S0} = 0.75$ and 0.5 give $A_{\rm B} = 1/7$ and $1/3$. Like the anti-greenhouse effect it warms the top relative to the bottom, but in addition the *whole* profile shifts cooler: the absorbed energy is reduced by $(1 - A_{\rm B})$.
- **Longwave scattering** does the opposite: thermal emission trying to escape is partly scattered back, and the atmosphere is warmer everywhere — the **scattering greenhouse**. Scattering in the infrared needs particles of a micron or larger.
- **Clouds and aerosols** can do all of these at once — greenhouse and anti-greenhouse, by absorption and by scattering. The effects may reinforce or cancel, which is why clouds complicate every interpretation.

Summary of the effects

effect	cause	effect on the T–P profile
greenhouse warming	more longwave opacity	lower layers warmer, upper cooler
collision-induced absorption	$\tau_{\mathrm{CIA}} \propto P^2$ (H_2, N_2)	deep layers warmer
anti-greenhouse cooling	more shortwave opacity	upper layers warmer, lower cooler; inversion if strong enough
higher Bond albedo	shortwave scattering	anti-greenhouse + whole profile cooler
scattering greenhouse	longwave scattering	warmer everywhere
interior heat	$T_{\mathrm{int}} > 0$	deep Milne “tail”

Each of these is a single parameter in the analytic model, which is why such models remain the standard tool for building intuition before a full radiative–convective calculation.

Convection basics

Convection transports energy by **bulk vertical motion**: heat flows from hot to cold regions with the moving gas. It is driven by the temperature gradient and enabled by gravity (buoyancy).

- A parcel heated from below becomes lighter than its surroundings and rises. At lower ambient pressure it expands and releases heat; now cooler, it sinks, is reheated, and the cycle repeats.
- Everyday examples: boiling water, lava lamps, the shimmering air above hot pavement.
- The Sun's **granulation**: bright centres are rising columns of hot gas, dark lanes are cooled gas sinking; cells are $\sim 1000\,\mathrm{km}$ across.

The dry adiabatic lapse rate

First law for a parcel, $\mathrm{d}U = \mathrm{d}Q - \mathrm{d}W$, with $\mathrm{d}Q = 0$ (adiabatic), $\mathrm{d}U = mc_v\,\mathrm{d}T = m(c_p - R_s)\,\mathrm{d}T$ and $\mathrm{d}W = P\,\mathrm{d}V$. The ideal gas law $PV = mR_sT$ gives $P\,\mathrm{d}V = mR_s\,\mathrm{d}T - (m/\rho)\,\mathrm{d}P$, so

$$\frac{\mathrm{d}T}{\mathrm{d}P} = \frac{1}{\rho c_p} \xrightarrow{\mathrm{d}P = -\rho g\,\mathrm{d}z} \boxed{\Gamma \equiv -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}}$$

- Γ is the **adiabatic lapse rate** (positive when temperature falls with height). A rising parcel shares the pressure of its surroundings but follows its own T and ρ.
- In optical depth, $\mathrm{d}T/\mathrm{d}\tau = g/(\kappa_m c_p)$.
- In pressure: $\mathrm{d}T/T = (R_s/c_p)\,\mathrm{d}P/P$, so along an adiabat

$$T \propto P^{R_s/c_p}, \qquad R_s/c_p = 2/7 \text{ for a diatomic ideal gas.}$$

On a log–log T–P diagram an adiabat is a straight line.

The moist adiabat

When a condensable vapour condenses in the rising parcel, its latent heat warms the parcel: $\mathrm{d}Q = -L\,\mathrm{d}m_s$, with m_s the vapour mass per unit mass of non-condensable gas and L the specific latent heat. Keeping $\mathrm{d}Q$ in the first law,

$$\boxed{\Gamma_{\mathrm{moist}} = \frac{g}{c_p}\,\frac{1}{1+(L/c_p)(\mathrm{d}m_s/\mathrm{d}T)}} \qquad (\mathrm{d}m_s/\mathrm{d}T > 0,\ \text{so}\ \Gamma_{\mathrm{moist}} < \Gamma).$$

- Earth's troposphere: dry $\Gamma \simeq 9.8\,\mathrm{K\,km^{-1}}$, moist $\simeq 5\,\mathrm{K\,km^{-1}}$; an average $6.5\,\mathrm{K\,km^{-1}}$ is typical since convection is partly dry, partly moist.
- The observed profile is *stable* against dry convection but *unstable* against moist convection: latent heat is what drives Earth's convection.
- Exoplanets with other temperatures condense other species (e.g. NH_3, CH_4, silicates, iron); the same formula applies with their L and saturation curves.

Conceptual overview

A radiative–convective model solves **three equations for three unknowns** — the temperature T, the pressure P, and the radiation field I_ν:

1. hydrostatic equilibrium, $\mathrm{d}P/\mathrm{d}\tau = g/\kappa_m$;
2. the transfer equation, $\mu\,\mathrm{d}I_\nu/\mathrm{d}\tau = I_\nu - S_\nu$;
3. radiative (and convective) equilibrium, $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$.

Procedure.

- Start from a guess, e.g. the grey profile with T- and P-dependent opacities.
- Solve the three equations numerically, applying temperature corrections until all are satisfied.
- Check every layer for convective stability. Where it is unstable, replace the profile by the adiabat and iterate again, so that the radiative part joins the adiabat smoothly.

A radiative-equilibrium atmosphere: inputs and subtleties

Outputs: $I(\tau, \mu, \nu)$, $T(\tau)$, $P(\tau)$ — the T–P profile and the emergent spectrum together.

Inputs: the interior temperature T_{int} (from evolution models, or free), the surface gravity g (known for transiting planets), the stellar irradiation, and the opacities — which need the composition, from chemical equilibrium or a photochemical model.

Subtleties:

- The equations are tightly coupled. A new $T(\tau)$ changes the chemistry, hence the number densities, the pressure, κ_ν and ε_ν, hence the radiation field everywhere.
- Scattering makes the transfer equation non-local: stellar photons can travel deep before they heat the gas, coupling the top of the atmosphere to the deep layers.
- Hence iteration from a good starting profile, typically a grey solution.

Radiative–convective equilibrium

Take the radiative-equilibrium profile and test each layer with the Schwarzschild criterion, $|\mathrm{d}T/\mathrm{d}z|_{\mathrm{rad}} > |\mathrm{d}T/\mathrm{d}z|_{\mathrm{ad}}$.

- Usually convection is assumed so efficient that, where it occurs, the gradient *is* the adiabat. The adiabat fixes only the slope; its level is set by matching the radiative temperature at the top of the convection zone.
- Where energy is shared, the net flux is split: $F_{\mathrm{rad}} + F_{\mathrm{conv}} = \sigma T_{\mathrm{int}}^4$ (deep layers, where the starlight has been absorbed).
- The convective flux is

$$F_{\mathrm{conv}} = \bar{v}\,\rho c_p\,\Delta T, \qquad \Delta T = \left[\left(-\frac{\mathrm{d}T}{\mathrm{d}z}\right) - \left(-\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{c}}\right]\Delta z,$$

  the excess heat of a parcel that travels Δz at mean speed $\bar{v}$ before merging with its surroundings.
- $\bar{v}$ and Δz come from **mixing-length theory**, a local 1D approximation from stellar physics: the mixing length $l \sim H$ is the distance a bubble rises before releasing its heat.

Seager's Eqs. (9.75)–(9.76) write the constant flux as $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$ and $\sigma T_{\mathrm{eff}}^4 + \mu_0 I_0$. The net (up minus down) flux that is constant with depth is $\sigma T_{\mathrm{int}}^4$; $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$ is the outgoing thermal flux at the top.

Is the deep atmosphere convective? Milne's solution and CIA

Write the Schwarzschild criterion in logarithmic form: the atmosphere is stable if

$$\frac{\partial \ln T}{\partial \ln P} \leq \nabla_{\rm ad} = \frac{R_s}{c_p} = \frac{2}{2+n_{\rm dof}},$$

with $n_{\rm dof}$ the degrees of freedom of the dominant gas: for a diatomic gas, $\nabla_{\rm ad} = 2/7$ (translation and rotation) or $1/4$ (with vibration). H_2 (giant planets) and N_2 (Earth, Titan) are diatomic.

Deep down, interior heat gives Milne's solution, $T^4 \propto \tau$:

- **Constant opacity:** $\tau \propto P$, so $T \propto P^{1/4}$ and $\partial \ln T/\partial \ln P \to 1/4 \leq 2/7$: **stable**, radiative.
- **CIA dominates:** $\kappa \propto P$, so $\tau \propto P^2$, $T \propto P^{1/2}$ and $\partial \ln T/\partial \ln P \to 1/2 > 2/7$: **unconditionally convective**.

The result does not depend on the strength of absorption, on scattering, or on gravity: when collision-induced absorption operates deep down, the interior of a gas giant must convect. Quantum-mechanical opacities, via Schwarzschild's criterion, decide a macroscopic flow.

Higher up, irradiation usually stabilizes the profile; but with a strong greenhouse (shortwave opacity much smaller than longwave), *detached* convective layers can appear between stable zones (Heng, problem 4.9.2).

Forward and inverse problems

- The **forward problem**: physics → T–P profile → spectrum, by iterating the coupled equations.
- The **inverse problem**: spectrum → T–P profile. For Solar System planets, high-quality spectra allow a fiducial profile to be inferred and then perturbed (in T and abundances) until the data are fit.
- Exoplanet spectra are not good enough to infer a unique fiducial profile. Instead, map out the range of T–P profiles and abundances the data allow.
- Full forward models are too slow to run millions of times, so Madhusudhan & Seager (2009) used a **parametric T–P profile**, running tens of thousands of models that satisfy hydrostatic equilibrium and global energy balance. The result constrains the T–P range, molecular abundances, albedo, day–night redistribution and effective temperature.

A parametric T–P profile

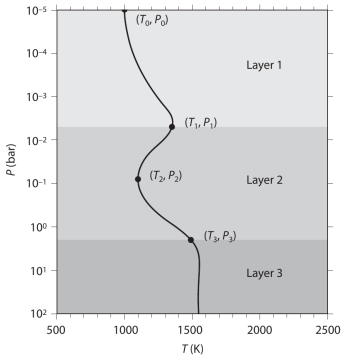


Figure 13: The parametric P–T profile, with a possible inversion in layer 2. Seager (2010), Fig. 9.8, after Madhusudhan & Seager (2009).

$$P = P_0\, \mathrm{e}^{\alpha(T-T_0)^{\beta}} \quad \text{in layers 1 and 2,} \qquad T = T_3 \text{ in layer 3,}$$

with $\beta = 1/2$ and continuity at the layer boundaries; with $P_0 = 10^{-5}$ bar this leaves **six free parameters**.

- **Above P_0:** transparent at all wavelengths; no spectral features.
- **Layer 1:** no inversion; heated from below, cooling upward.
- **Layer 2:** where most features form; radiative (and perhaps dynamical) processes; an **inversion** may form with strong irradiation and absorbers.
- **Layer 3:** high optical depth, radiative diffusion, nearly isothermal — heated by the interior, not the star. Below it, convection.
- Cooler planets: no layer 3, layer 2 extends deeper, and the radiative–convective boundary is higher (an adiabat in layers 2–3).

Take-home messages

Approximate solutions. Rosseland (deep layers): $F = -(16\sigma T^3/3\alpha_R)\,\mathrm{d}T/\mathrm{d}z$ with the harmonic mean α_R, which with constant flux gives Milne's $T^4 = \frac{3}{4}T_{\rm int}^4(\tau_R + C)$. Eddington: $K = J/3$, which in a grey atmosphere gives $J = 3H(\tau + \frac{2}{3})$, $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$ and the limb-darkening law $I(\mu)/I(1) = \frac{3}{5}(\mu + \frac{2}{3})$.

Structure. Solar System atmospheres share a troposphere, tropopause near 0.1 bar and (UV-heated) stratosphere. Hydrostatic equilibrium, $\mathrm{d}P/\mathrm{d}z = -\rho g$, with the ideal gas law gives $P = P_0 \mathrm{e}^{-z/H}$, $H = kT/\mu_m m_{\rm H} g$: 8 km for Earth, hundreds of km for hot Jupiters; μ_m matters as much as T.

Greenhouse models. One opaque layer: $T_s = 2^{1/4}T_e$; leaky: $T_s = [2/(2-\alpha)]^{1/4}T_e$; n layers: $(n+1)^{1/4}T_e$ (Venus needs ~ 100).

Transport. Convection when $|\mathrm{d}T/\mathrm{d}z|_{\rm rad} > g/c_p$: high opacity, large flux, or large c_p. Deep layers and surfaces under thin atmospheres convect; inversions are stable.

Radiative equilibrium. $\int \kappa_\nu(J_\nu - S_\nu)\mathrm{d}\nu = 0$. Grey, heated from below: $T^4 = \frac{3}{4}T_{\rm int}^4(\tau + \frac{2}{3})$. Heated from above too: two bands with $\gamma = \kappa_{\rm vis}/\kappa_{\rm ir}$; an inversion forms when $\gamma > \sqrt{3}\,\mu_0$.

Greenhouse and anti-greenhouse. $A_{\rm B} = (1-\beta_{S0})/(1+\beta_{S0})$; starlight is deposited near $P_D \approx 0.63\, g\beta_{S0}/\kappa_S$. Longwave opacity or CIA warms the deep layers; shortwave opacity warms the top and can invert the profile; albedo cools; infrared scattering warms. With CIA the deep Milne profile ($T \propto P^{1/2}$) must convect.

Adiabat and models. $\Gamma = g/c_p$ (9.8 K/km dry on Earth), smaller when vapour condenses; $T \propto P^{R_s/c_p}$. Full models solve hydrostatic, transfer and radiative–convective equilibrium together by iteration; retrievals use parametric T–P profiles.

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